

Taylor Howard

Senior Data Engineer

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Lake Mary, FL

SUMMARY

Senior Data Engineer with 10+ years of experience designing scalable **Azure** Data Lakehouse architectures and enterprise **ETL/ELT** pipelines across healthcare and consulting environments. Strong expertise in **Azure Data Factory (ADF)**, **Azure Databricks**, **ADLS Gen2**, **Delta Lake**, **PySpark**, **CDC ingestion frameworks**, and **Dimensional Modeling (Kimball methodology)** supporting multi-terabyte workloads. Experienced in near real-time ingestion, production reliability management, and hybrid-cloud integration. Proven ability to deliver high-performance, governance-aligned data platforms processing 100M+ records with strong compliance and operational controls.

SKILLS

Cloud Platforms:

Microsoft Azure (ADF, Databricks, Synapse, ADLS Gen2, Event Hub, Key Vault, Azure Monitor)
Snowflake (ELT, analytics warehouse, virtual warehouses), AWS (S3, Glue, IAM)

Data Engineering & Lakehouse:

ETL/ELT pipelines, Medallion Architecture, Delta Lake, PySpark, SparkSQL, CDC, Incremental ingestion, Schema evolution, Partition optimization, Z-ORDER

Streaming & Messaging:

Spark Structured Streaming, Azure Event Hub

DevOps & Automation:

CI/CD (Azure DevOps, GitHub Actions), Docker, Terraform (collaborative IaC), Git

Data Modeling & Governance:

Dimensional Modeling (Kimball), Star Schema, Data Marts, Data validation frameworks, Data retention policies

EXPERIENCE

Senior Data Engineer

Cognizant | Jan 2022 – Present

- Served as technical owner for ingestion and transformation layers, leading design and implementation of scalable Azure Data Lakehouse solutions.
- Architected enterprise-scale Azure-based platform using ADF, Databricks, ADLS Gen2, and Delta Lake, processing ~4 TB daily (~120M+ records across claims, enrollment, and provider datasets).
- Designed and implemented end-to-end ETL and ELT pipelines using Medallion Architecture (Bronze/Silver/Gold layers) for standardized ingestion and transformation.
- Reduced batch runtime from 6 hours to ~3.5 hours through Spark partition optimization and execution tuning.
- Built near real-time ingestion pipelines using Azure Event Hub and Spark Structured Streaming, reducing reporting latency to under one hour.

- Implemented CDC and incremental load strategies reducing reliance on full refresh processing.
- Applied Delta Lake optimization techniques (OPTIMIZE, Z-ORDER, compaction) improving large-scale analytical query performance.
- Developed ingestion pipelines consuming REST APIs and semi-structured JSON datasets with schema evolution handling.
- Designed idempotent pipeline logic and controlled historical backfill frameworks for safe reprocessing of legacy datasets.
- Maintained production reliability exceeding 99% successful pipeline execution across regulatory and operational workloads.
- Led root cause analysis for production incidents and implemented monitoring safeguards to prevent recurrence.
- Automated CI/CD deployments using Azure DevOps improving release consistency and rollback safety.
- Collaborated with cloud engineering teams on Terraform-based Infrastructure-as-Code provisioning.
- Implemented monitoring and alerting using Azure Monitor and Log Analytics improving operational visibility.
- Partnered with enterprise data governance teams to align ingestion frameworks with data lineage, retention, and access control standards.
- Supported limited Snowflake usage as downstream analytics warehouse for curated datasets and reporting workloads.
- Mentored junior engineers and promoted Spark performance tuning best practices.

Senior Data Engineer

UnitedHealth Group | Jan 2016 – Nov 2021

- Designed and maintained enterprise Data Warehouse and Azure Lakehouse environments supporting 10M+ member records and high-volume claims datasets.
- Built scalable ETL pipelines using Azure Data Factory and Databricks integrating claims, enrollment, and provider systems.
- Reduced ETL processing time by ~40% through Spark partition refinement and workload optimization.
- Engineered ELT transformation workflows supporting schema evolution and reliable analytics delivery.
- Developed dimensional data models (Star Schema) supporting CMS, HEDIS, financial, and operational analytics.
- Built curated data marts enabling actuarial forecasting and executive reporting.
- Implemented Python-based data validation and reconciliation frameworks improving production data quality.
- Designed incremental ingestion and change-tracking mechanisms reducing refresh windows.
- Optimized complex SQL and Spark transformations for large healthcare datasets.
- Integrated external vendor datasets via secure API ingestion pipelines.
- Implemented workload isolation strategies separating production and ad-hoc analytics workloads.
- Participated in CI/CD pipeline implementation and Git-based release management processes.
- Maintained consistent production pipeline reliability across enterprise reporting systems.
- Collaborated with actuarial, analytics, and product teams to align data models with evolving business rules.
- Contributed to enterprise data governance initiatives including audit readiness and data retention alignment.
- Supported limited hybrid ingestion scenarios leveraging AWS S3 for secure partner data exchange.

Data Engineer

DaVita | Sep 2012 – Jan 2016

- Developed ETL workflows integrating clinical and operational datasets into centralized warehouse environments.
- Designed Star Schema data models supporting executive and operational reporting.
- Tuned SQL queries reducing report generation time.
- Implemented automated job scheduling improving pipeline stability and monitoring visibility.
- Built Power BI dashboards supporting clinical analytics and leadership reporting.
- Implemented validation routines improving dataset integrity and reporting consistency.
- Supported HIPAA-compliant reporting processes and secure data handling.
- Collaborated with cross-functional teams to gather reporting requirements and deliver analytics-ready datasets.

EDUCATION

University of Columbia | New York, NY | Aug 2008 – Jun 2012

Bachelor | Computer Science